

CONEXUS NETWORK SOLUTIONS PVT LTD

Pulse

by Conexus — People & HR Platform

Product Specification & System Architecture — v3.1

Confidential | Internal Use Only | 2025

STACK: Laravel 11 (PHP 8.3) + MySQL 8 + Livewire 3 + Blade + Tailwind CSS
YEAR: Conexus Annual Year runs July 1 – June 30
EMPLOYEES: All India-based. Hybrid model (office + WFH).
SHIFTS: IT Shift 10:30–7:30 IST | UK Sales Shift 1:00–10:00 PM IST | Grace: 5 mins
LEAVE: 18 days total = 12 Casual/Sick + 6 Mandatory December. No lapse on any leave type. Encashment on approval.
OT: Pre-approved by manager. Fixed rate ₹100/hr.
SALARY: Two cycles: 1–31 or 21–20. Configurable per employee.
NOTIFY: In-app notification centre (bell icon). Email for critical actions only.
INTEGRATIONS: None (no Slack, no ClickUp, no Google Calendar).

Section 1 — Product Overview

1.1 Purpose

Pulse is Conexus's lean, self-hosted internal people platform built on Laravel. It centralises HR operations for a hybrid India-based team working across two shifts. Pulse eliminates manual HR administration without introducing external dependencies. The Conexus financial year runs July 1 to June 30; all leave balances, performance cycles, and reporting reset on this basis.

1.2 What Is Out of Scope

- OUT OF SCOPE — deliberately excluded to keep the system lean:
- Third-party payroll platforms (Keka, greytHR)
 - Slack, ClickUp, or any project management integrations
 - Google Calendar integration
 - Recruitment / ATS module
 - Native mobile app — responsive web (PWA-ready) only
 - UK-specific statutory payroll (PAYE, NI) — all staff are India-based
 - Bulk email blasts — email used only for critical actions (e.g. payslip delivery)

1.3 Core Modules

Module	Purpose
Employee Management	Profiles, departments, org structure, document storage
Attendance	Clock in/out, break/pause codes, late flags, regularisation requests
Leave Management	CSL + MDL + Comp Off, encashment requests, approval workflow
Overtime	Pre-approval request flow, ₹100/hr rate, monthly OT summary
Compensation & Finance	Salary records, two pay cycles, OT pay, incentives, reimbursements, payslips
In-App Notifications	Bell icon notification centre with read/unread state
Performance	Quarterly KPI reviews aligned to QBR cycle
Onboarding / Offboarding	Checklists, equipment log, exit process
Document Management	Policies, contracts, payslips, acknowledgements

1.4 User Roles

Role	Who	Access Level
Super Admin	Mazhar	Full system access
HR Admin	Shivani	All HR modules except system config

Role	Who	Access Level
Department Heads	Rustom, Nick, Nikia	Own department data, approvals, reports
Manager	Team leads	Team attendance, leave/OT approvals, reviews
Finance	Emad	Compensation module + read-only employee/attendance summary
Employee	All staff	Self-service: attendance, leave, OT requests, payslips, notifications

Section 2 — System Architecture

2.1 Stack

Layer	Technology	Notes
Backend	Laravel 11 (PHP 8.3)	MVC, Eloquent ORM, queues, scheduling, policies
Frontend	Livewire 3 + Blade	Reactive UI server-rendered; no separate JS framework
CSS	Tailwind CSS 3	Utility-first; no custom stylesheet sprawl
Database	MySQL 8	Relational integrity for all HR data
Auth	Laravel Breeze	Session-based auth, password reset, MFA optional
Email	Laravel Mail + SMTP	Used only for payslip delivery and critical HR actions
Notifications	Laravel Notifications (database channel)	In-app bell icon; no email spam
Queue	Laravel Queue (database driver)	Async dispatch; no Redis needed at this scale
Scheduler	Laravel Scheduler (cron)	Attendance checks, reminders, monthly summaries
File Storage	Laravel Storage (local disk → S3-ready)	Documents, payslips; swap to S3 via single .env change
PDF	barryvdh/laravel-dompdf	Payslip and letter generation
Hosting	VPS — Ubuntu 22 + Nginx + PHP-FPM	~₹1,500–2,000/mo on DigitalOcean or AWS EC2

2.2 Notification Architecture (In-App)

All system notifications use Laravel's database notification channel. A notifications table stores each alert with a read/unread flag per user. The bell icon in the nav bar shows an unread count badge. Employees click the bell to see a dropdown of recent notifications with timestamps. A 'Mark all read' action is provided.

EMAIL is sent ONLY for:

- Payslip issued (payslip PDF attached)
- Account created / password reset
- Final settlement on offboarding

EVERYTHING ELSE uses in-app notifications only:

- Leave request submitted / approved / rejected
- OT request submitted / approved / rejected

- Attendance regularisation submitted / approved / rejected
- Leave encashment request submitted / approved
- Performance review due / submitted
- Document to acknowledge
- Payroll ready for Finance approval
- Probation and check-in reminders (to manager)
- Excess break flag
- Missing clock-out reminder

2.3 Application Structure

app/Http/Controllers/	— one controller per module
app/Http/Livewire/	— Livewire components (forms, tables, dashboards)
app/Models/	— Eloquent models
app/Services/ AttendanceService)	— business logic (LeaveService, PayrollService, OTService,
app/Notifications/	— all notification classes (database channel)
app/Policies/	— RBAC via Laravel Gates and Policies
database/migrations/	— all schema versioned
resources/views/	— Blade templates
routes/web.php	— role-grouped route definitions
app/Console/Commands/	— scheduled artisan commands

2.4 Authentication & RBAC

- Laravel Breeze for login / logout / password reset
- Role stored as ENUM on users table: super_admin, hr_admin, director, manager, finance, employee
- Laravel Gates and Policies enforce access at controller and Blade layer
- Route groups protected by role middleware: Route::middleware(['auth', 'role:manager,director'])
- 5-minute session grace on clock-in timestamps handled at the Service layer, not as a route guard

Section 3 — Core Modules

3.1 Employee Management

- Employee profile: name, email, phone, emergency contact, date of birth, gender, address, photo
- Employment record: designation, department, reporting manager, shift assignment, work mode, employment type, start date, probation end date, status
- Salary cycle assignment: configurable per employee (Cycle A: 1–31 or Cycle B: 21–20)
- Org chart: hierarchy view rendered from manager_id relationships
- Document storage: contracts, ID proofs, signed policies linked to employee record
- Employee lifecycle states: active, probation, notice_period, inactive (archived, never deleted)

Table	Key Columns
employees	id, user_id, emp_code, full_name, email, phone, dob, gender, address, emergency_contact, photo, status
employment_details	id, employee_id, designation, department_id, manager_id, shift (it/uk), work_mode (office/wfh/hybrid), employment_type, salary_cycle (A/B), start_date, probation_end_date
departments	id, name, head_employee_id, description

3.2 Attendance Management

Shifts & Grace Period

Shift	Hours (IST)	Duration	Grace	Who
IT Shift	10:30 AM – 7:30 PM	9 hrs	5 mins	Dev, Ops, Marketing
UK Sales Shift	1:00 PM – 10:00 PM	9 hrs	5 mins	UK Sales / BD team

Grace period: 5 minutes applies to both shifts.

IT Shift — clock-in up to 10:35 AM is on time. 10:36 AM onwards is flagged as Late.

UK Sales Shift — clock-in up to 1:05 PM is on time. 1:06 PM onwards is flagged as Late.

OT calculation: hours beyond 9 hrs are recorded as OT hours only if an approved OT request exists for that day.

Clock In / Out & Break Features

- Clock in / out via web dashboard button; work mode (Office/WFH) selected at clock-in
- IP address logged at clock-in as a soft audit trail (no hard block)
- Break / Pause code: employee taps 'Start Break' and 'End Break' — repeatable any number of times
- Each break segment recorded with start time, end time, and duration in minutes
- Total break time shown informational; hours NOT deducted (total_hours = clock_out – clock_in)
- Excess break flag raised if total break time exceeds 60 minutes in a day

- Missing clock-out: flagged if no clock-out recorded by shift end + 1 hour; in-app notification sent to employee

Attendance Regularisation

Employees can submit a regularisation request if they missed a clock-in or clock-out, or clocked incorrect hours due to a system or human error. The request goes to their manager for approval.

1. Employee submits regularisation request: date, correct clock-in time, correct clock-out time, reason
2. In-app notification sent to manager
3. Manager reviews and approves or rejects with comment
4. On approval: attendance_log for that date is updated; original values stored in audit trail
5. Employee notified in-app of outcome

Table	Key Columns
attendance_logs	id, employee_id, date, clock_in, clock_out, total_hours, total_break_minutes, ot_hours, work_mode, late_flag, excess_break_flag, missing_checkout_flag, ip_address, regularised (bool), notes
break_logs	id, attendance_log_id, employee_id, break_start, break_end, duration_minutes
attendance_regularisations	id, employee_id, date, requested_clock_in, requested_clock_out, reason, status, reviewed_by, reviewed_at, manager_comment
shift_settings	id, shift_name, start_time, end_time, standard_hours, grace_minutes (5), max_break_minutes (60), ot_threshold_hours (9)

3.3 Leave Management

Leave Policy

CONEXUS ANNUAL YEAR: July 1 – June 30

TOTAL ENTITLEMENT: 18 days per year

- 12 days — Casual / Sick Leave (CSL)
- 6 days — Mandatory December Leave (MDL)

NO LAPSE POLICY: unused CSL balance does not expire at year end.

ENCASHMENT: employee may request encashment of unused CSL balance; subject to Director/HR approval.

Leave Type	Code	Days	Half-Day?	Carry Fwd?	Encashable?	Notes
Casual / Sick Leave	CSL	12	Yes	Yes (no lapse)	Yes — on approval	Combined pool; usable any time July–June
Mandatory December Leave	MDL	6	No	N/A	No	Pre-blocked company shutdown days in December

Leave Type	Code	Days	Half-Day?	Carry Fwd?	Encashable?	Notes
Comp Off	CO	Earned	No	No lapse	No	Earned by working on MDL day or UK public holiday. Carries forward indefinitely.
Public Holidays	—	UK calendar	—	—	—	Non-working days; no leave consumed

MANDATORY DECEMBER LEAVE:

6 specific days each December are designated as company shutdown and pre-blocked in the system.

Employees cannot apply for other leave types on these days.

If an employee works on an MDL day (e.g. for a client deadline), they earn 1 Comp Off credit.

Comp Off credits carry forward indefinitely — there is no expiry date. This is consistent with the no-lapse policy.

NO LAPSE ON CSL:

Unused CSL days carry forward indefinitely within the employee's tenure.

There is no automatic annual reset or expiry of CSL balance.

Encashment of accumulated CSL requires a formal request and Director/HR Admin approval.

Leave Approval Workflow

6. Employee submits leave request: type, dates, half-day flag, optional reason
7. System validates: sufficient balance, no overlap with approved leave, not a public holiday or MDL day
8. In-app notification sent to reporting manager
9. Manager approves or rejects with optional comment
10. If no action in 24 hours: escalation notification sent to HR Admin
11. Employee notified in-app of outcome; balance updated on approval

Leave Encashment Workflow

12. Employee submits encashment request: number of CSL days to encash, reason
13. System checks available CSL balance; blocks if insufficient
14. In-app notification to Director and HR Admin
15. Director or HR Admin approves or rejects
16. On approval: CSL balance reduced; encashment amount fed into current month's payroll as a line item
17. Employee notified in-app

Table	Key Columns
leave_types	id, name, code, days_per_year, allow_half_day, carry_forward, encashable

Table	Key Columns
leave_balances	id, employee_id, leave_type_id, financial_year, total_days, used_days, pending_days, comp_off_credits
leave_requests	id, employee_id, leave_type_id, start_date, end_date, half_day, reason, status, approved_by, approved_at, manager_comment
leave_encashment_requests	id, employee_id, days_requested, encashment_amount, reason, status, approved_by, approved_at, payroll_month_id
public_holidays	id, date, name, year
december_mandatory_days	id, date, year, is_comp_off_eligible

3.4 Overtime Management

OT POLICY:

- Overtime is only recorded and paid if it has been pre-approved by the employee's manager.
- Flat rate: ₹100 per hour (no differential for weekdays vs weekends at this stage).
- OT hours on a given day are only counted if an approved OT request exists for that date.
- Attendance clock-out time is the source of truth for actual OT hours worked.
- OT hours = actual hours worked – 9 standard hours (only if approved OT request covers the date).

OT Request Workflow

18. Employee submits OT request before working late: date, estimated hours, reason/project context
19. In-app notification sent to manager
20. Manager approves or rejects (with comment)
21. Employee notified in-app
22. On the approved date: attendance clock-out is recorded as usual; system calculates OT hours = total_hours – 9
23. OT hours and amount (₹100/hr) auto-populated in the monthly OT summary for Finance review
24. Finance verifies OT hours before payslip is generated

Database Tables

Table	Key Columns
ot_requests	id, employee_id, request_date, estimated_hours, reason, status (pending/approved/rejected), approved_by, approved_at, manager_comment
overtime_records	id, employee_id, attendance_log_id, ot_request_id, date, ot_hours_worked, ot_rate (100), ot_amount, month, year, verified_by

3.5 Compensation & Finance Module

This module manages salary records, pay cycle processing, OT pay, incentives, and reimbursements. It produces data for Finance to review and approve. No bank transfers or statutory filings are handled within the system.

Salary Cycles

Cycle	Period	Pay Date	Assignment
Cycle A	1st – 31st of month	1st of following month	Configurable per employee (salary_cycle = A)
Cycle B	21st – 20th of month	21st of following month	Configurable per employee (salary_cycle = B)

Salary cycle is set on the employment_details record for each employee.
 The system generates separate payroll runs for Cycle A and Cycle B each month.
 An employee can be moved between cycles by HR Admin; the change takes effect from the next cycle start.

Payslip Calculation

- Gross = basic + HRA + special allowance
- Deductions = any agreed deductions (advance recovery, etc.)
- OT pay = approved OT hours × ₹100
- Incentives = approved incentive amounts for the month
- Reimbursements = approved claims for the period
- Leave encashment = approved encashment amount if applicable
- Net pay = gross + OT + incentives + reimbursements + encashment – deductions

Payroll Approval Flow

25. HR Admin locks attendance and leave data for the cycle period
26. System generates draft payslips for all employees in the cycle
27. Finance (Emad) receives in-app notification: payroll ready for review
28. Finance reviews OT hours, incentives, reimbursements; adjusts if needed
29. Finance approves the payroll run
30. Payslips auto-generated as PDF (dompdf); delivered to employees via email (one of the few email triggers)
31. Payroll run archived; salary history updated

Incentives

- Ad hoc incentive entries: type, amount, reason, month — created by HR Admin or Director
- Types: performance bonus, sales commission, referral, project completion
- Requires Director approval before inclusion in payslip

Reimbursements

- Employee submits claim: category, amount, expense date, receipt upload (PDF/image)
- Categories: travel, internet (WFH), meal, medical, other
- Manager approves; Finance includes in payslip or processes as separate payment

Database Tables

Table	Key Columns
salary_structures	id, employee_id, basic, hra, special_allowance, gross_ctc, effective_from, effective_to, updated_by
payroll_runs	id, cycle (A/B), month, year, status (draft/pending_approval/approved), locked_by, approved_by, approved_at
payslips	id, employee_id, payroll_run_id, basic, hra, allowances, ot_amount, incentives_total, reimbursements_total, encashment_amount, gross, deductions, net_pay, pdf_path, emailed_at
incentives	id, employee_id, type, amount, reason, month, year, status, approved_by
reimbursements	id, employee_id, category, amount, expense_date, receipt_path, month, year, status, approved_by, notes

3.6 In-App Notification Centre

All system alerts are delivered through a persistent notification centre accessible via a bell icon in the top navigation bar. The unread count badge updates in real-time via Livewire polling (every 30 seconds).

Notification Events

Event	Notified Role(s)
Leave request submitted	Manager
Leave approved / rejected	Employee
Leave escalation (24 hrs no action)	HR Admin
OT request submitted	Manager
OT request approved / rejected	Employee
Attendance regularisation submitted	Manager
Attendance regularisation approved / rejected	Employee
Leave encashment request submitted	Director + HR Admin
Leave encashment approved / rejected	Employee
Excess break flag (>60 mins)	Employee + Manager
Missing clock-out reminder	Employee
Payroll ready for approval	Finance
Payslip issued (in-app + email)	Employee
Performance review due (7 days before quarter end)	Employee + Manager
Document requires acknowledgement	Employee
Document expiring in 30 days	HR Admin

Event	Notified Role(s)
Probation review due (10 days before)	Manager + HR Admin
30-day new hire check-in due	Manager
Incentive approved and added to payslip	Employee
Reimbursement approved / rejected	Employee

Database Table

Table	Key Columns
notifications	id, type, notifiable_type, notifiable_id, data (JSON: message, action_url, related_model, related_id), read_at, created_at
Uses Laravel's built-in notifications table (generated by php artisan notifications:table). read_at is null for unread; set to current timestamp when employee opens the notification. Bell icon badge = COUNT(*) WHERE notifiable_id = auth()->id() AND read_at IS NULL. Notifications older than 90 days are pruned by a scheduled artisan command.	

3.7 Performance Management

- Quarterly review cycle aligned to Conexus year: Q1 Jul–Sep, Q2 Oct–Dec, Q3 Jan–Mar, Q4 Apr–Jun
- KPIs defined per department at the start of each quarter by Directors / HR Admin
- Employee self-assessment: rates and comments on each KPI before manager review
- Manager review: rates each KPI, adds overall feedback, marks promotion flag if applicable
- Final rating: composite score from KPI weights
- Full review history visible to employee and manager

Department	Sample KPIs
Operations (Rustom, Nick, Nikita)	Project delivery on time %, code quality score, client satisfaction, skill growth
BD / Sales (Nick)	Revenue vs target, new clients, retention rate, invoice collection rate
Marketing / ED (Mazhar)	Campaign ROI, SEO movement, deliverable completion, team satisfaction
Cross-Dept	QBR goal completion %, process improvements, collaboration rating

Table	Key Columns
kpis	id, department_id, name, description, weight, quarter, financial_year
performance_reviews	id, employee_id, reviewer_id, quarter, financial_year, status, overall_rating, manager_feedback, promotion_flag
kpi_scores	id, review_id, kpi_id, self_rating, self_comment, manager_rating, manager_comment

Table	Key Columns
goals	id, employee_id, quarter, financial_year, description, status, completion_note

3.8 Onboarding & Offboarding

Onboarding Checklist

#	Task	Owner
1	Create Pulse login account and assign role	HR Admin
2	Set shift, work mode, salary cycle in system	HR Admin
3	Set up company email account	HR Admin / IT
4	Issue and log equipment (laptop, access card, etc.)	HR Admin / IT
5	Send employment contract for digital signature and upload to docs	HR Admin
6	Share policy documents; mark as requiring acknowledgement	HR Admin
7	Add to department and reporting structure	HR Admin
8	Assign buddy and schedule Day 1 orientation	Manager
9	30-day check-in (auto-notification to manager)	Manager
10	90-day probation review (auto-notification to manager + HR Admin)	Manager + HR Admin

Offboarding Checklist

#	Task	Owner
1	Log exit: type (resignation/termination), last working day	HR Admin
2	Recover all equipment; log return and condition	HR Admin / IT
3	Revoke Pulse access on last working day	HR Admin
4	Revoke email and tool access	HR Admin / IT
5	Calculate full & final settlement (fed into Compensation module)	Finance
6	Generate experience letter	HR Admin
7	Conduct exit interview; record notes in system	HR Admin / Director
8	Archive employee record (status → inactive; never deleted)	HR Admin

Table	Key Columns
onboarding_tasks	id, employee_id, task_name, owner_role, due_date, completed_at, completed_by, notes
equipment_log	id, employee_id, item_name, serial_no, issued_at, returned_at, condition_on_return

Table	Key Columns
exit_records	id, employee_id, exit_date, exit_type, exit_reason, interview_notes, assets_returned, final_settlement_status, archived_at

3.9 Document Management

- Company documents: handbook, policies, code of conduct — visible to all staff
- Employee documents: contracts, appraisal letters, payslips — scoped to individual + HR Admin + Finance
- Version control: re-upload creates a new version; prior versions retained
- Policy acknowledgement: HR Admin can flag a document as requiring employee confirmation
- Expiry tracking: documents with an expiry date trigger an in-app notification to HR Admin 30 days before

Table	Key Columns
documents	id, name, category (company/employee), employee_id (nullable), file_path, version, uploaded_by, uploaded_at, expiry_date, requires_acknowledgement
document_acknowledgements	id, document_id, employee_id, acknowledged_at

Section 4 — Role-Based Access Control

Module	Super Admin	HR Admin	Director	Manager	Finance	Employee
Employee Profiles	Full CRUD	Full CRUD	View dept	View team	View basic	View own
Attendance	Full	Full	View dept	View + approve regularise	Summary only	Own + regularise
Leave	Full + override	Full	Approve dept	Approve team	Summary	Own only
OT Requests	Full	Full	View dept	Approve team	OT summary	Submit own
Compensation	Full	Configure	View dept cost	No access	Full access	Own payslips
Notifications	All	All	Own + team	Own + team	Own	Own only
Performance	Full	Full	Review dept	Review team	No access	Own reviews
Onboarding/Exit	Full	Full	View dept	Team tasks	Exit settlement	Own checklist
Documents (HR)	Full	Full	Policies only	Policies only	Policies only	Policies only
Documents (Payslips)	Full	Full	No access	No access	Full access	Own only
Reports	All	All HR	Dept dash	Team dash	Finance reports	Own summary
System Settings	Full	Partial	None	None	None	None

4.1 Special Role Notes

Shivani (HR Admin): HR Admin role with Manager-level read access across all departments. Implemented via a secondary permissions flag.

Emad (Finance): Finance role with direct payroll approval access. Can view attendance and leave summaries for cost calculations.

Nikita (Department Head): Department Head role scoped to UK Sales shift team only.

Section 5 — Dashboard Design

5.1 Executive Dashboard (Mazhar)

- Headcount: total active employees, by department, by shift, by work mode
- Today's attendance: % clocked in, WFH vs office, any absences without leave
- Leave overview: pending approvals company-wide, on-leave today
- Payroll status: current cycle progress, pending Finance approval
- Performance: QBR review completion %, average rating by department
- Alerts panel: expiring documents, overdue onboarding tasks, probation reviews due

5.2 Operations / BD Dashboard (Rustom / Nick / Nikita — Department Head)

- Team attendance: daily clock-in status for their department
- Leave calendar: visual monthly team leave calendar
- Pending approvals: leave and OT requests awaiting action (with age indicator)
- OT summary: team OT hours and amount for current month
- Performance: team KPI completion status, outstanding reviews

5.3 Finance Dashboard (Emad)

- Payroll approval queue: Cycle A and Cycle B drafts ready for review
- Monthly payroll summary: gross, OT, incentives, reimbursements, net — by department and total
- OT records: all approved OT hours for current cycle by employee
- Incentives: pending and approved incentives for the month
- Reimbursements: claims approved and awaiting payment
- Leave encashment: approved encashment requests to include in payslip

5.4 Employee Self-Service Dashboard

- Attendance widget: clock-in/out button, work mode toggle, Start/End Break button, today's hours and break time
- This month: attendance summary, late flags, excess break flags, leave taken
- Leave balances: CSL remaining, MDL days this December, Comp Off credits
- OT requests: pending / approved requests; submit new OT request
- Payslips: last 12 months with PDF download
- Notifications: recent unread alerts with action links
- Team calendar: colleagues on leave this week (name + dates only)

Section 6 — Database Schema (MySQL)

Table	Purpose	Key FK Relationships
users	Auth accounts	—
employees	Core employee record	users.id
employment_details	Role, dept, shift, cycle	employees.id, departments.id
departments	Org structure	employees.id (head)
shift_settings	Shift config + grace	—
attendance_logs	Daily clock records	employees.id
break_logs	Break segments per day	attendance_logs.id, employees.id
attendance_regularisations	Correction requests	employees.id
leave_types	Leave type definitions	—
leave_balances	Balance per employee/type/year	employees.id, leave_types.id
leave_requests	Leave submissions	employees.id, leave_types.id
leave_encashment_requests	Encashment requests	employees.id
public_holidays	UK holiday calendar	—
december_mandatory_days	MDL shutdown days	—
ot_requests	Pre-approval for overtime	employees.id
overtime_records	Actual OT worked + pay	employees.id, ot_requests.id, attendance_logs.id
salary_structures	Pay components history	employees.id
payroll_runs	Monthly batch per cycle	—
payslips	Individual payslip	employees.id, payroll_runs.id
incentives	Bonus/incentive entries	employees.id
reimbursements	Expense claims	employees.id
notifications	In-app notification store	employees.id (notifiable)
kpis	KPI definitions	departments.id
performance_reviews	Quarterly reviews	employees.id
kpi_scores	Per-KPI ratings	performance_reviews.id, kpis.id
goals	Quarterly goals	employees.id
onboarding_tasks	New hire checklist	employees.id
equipment_log	Asset allocation	employees.id

Table	Purpose	Key FK Relationships
exit_records	Offboarding data	employees.id
documents	All HR documents	employees.id (nullable)
document_acknowledgements	Read confirmations	documents.id, employees.id

6.1 Indexing Strategy

- employees: idx on user_id, emp_code, status
- attendance_logs: composite idx on (employee_id, date); idx on late_flag, excess_break_flag
- break_logs: idx on attendance_log_id, employee_id
- attendance_regularisations: idx on (employee_id, status)
- leave_requests: idx on (employee_id, status); idx on start_date
- leave_balances: composite idx on (employee_id, leave_type_id, financial_year)
- ot_requests: idx on (employee_id, request_date, status)
- overtime_records: idx on (employee_id, month, year)
- payslips: idx on (employee_id, payroll_run_id)
- notifications: idx on (notifiable_id, read_at); prune records older than 90 days
- performance_reviews: idx on (employee_id, quarter, financial_year)

Section 7 — Scheduled Jobs (Laravel Scheduler)

Add to server crontab: `* * * * * cd /var/www/pulse && php artisan schedule:run >> /dev/null 2>&1`

Command	Schedule	Action
<code>pulse:check-missing-checkouts</code>	Daily at 21:00 IST	Flag attendance_logs with no clock-out; send in-app notification to employee
<code>pulse:check-late-arrivals</code>	Daily at 10:45 + 13:15 IST	Confirm late flags for each shift; notify manager via in-app
<code>pulse:check-excess-breaks</code>	Daily at 20:00 IST	Sum break_logs per employee; flag and notify if > 60 mins
<code>pulse:check-leave-escalations</code>	Hourly	Find leave requests pending > 24 hrs; notify HR Admin in-app
<code>pulse:check-ot-escalations</code>	Hourly	Find OT requests pending > 24 hrs; notify HR Admin in-app
<code>pulse:check-document-expiry</code>	Daily at 08:00	In-app notification to HR Admin for docs expiring in 30 days
<code>pulse:check-probation-due</code>	Daily at 08:00	Notify manager + HR Admin 10 days before probation end
<code>pulse:check-newhire-checkin</code>	Daily at 08:00	Notify manager when employee hits 30-day mark
<code>pulse:check-review-reminders</code>	Weekly on Monday at 09:00	Notify employees + managers if QBR review due within 7 days
<code>pulse:prune-notifications</code>	Weekly on Sunday	Delete notifications older than 90 days
<code>pulse:generate-attendance-summary</code>	1st of each month at 01:00	Generate previous month attendance summary report per employee

Section 8 — Implementation Roadmap

Phase	Duration	Deliverables
Phase 1 — Core Loop	6–8 weeks	Auth + RBAC, Employee Management, Attendance (clock in/out, breaks, regularisation), Leave (CSL + MDL + Comp Off + encashment), In-app notification centre, Basic dashboards (executive + employee self-service)
Phase 2 — Finance	4–5 weeks	OT request + approval flow, Compensation module (salary cycles A+B, OT pay at ₹100/hr, incentives, reimbursements), Payslip PDF generation, Finance dashboard, Payroll approval workflow
Phase 3 — People Ops	4–5 weeks	Performance reviews + KPIs (aligned to July–June year), Onboarding/offboarding checklists, Document management + acknowledgements, Remaining dashboards (Director, Finance, Manager)
Phase 4 — Polish	2–3 weeks	Reports export (PDF/Excel), Audit log viewer for HR Admin, System settings UI (shift config, cycle assignment, MDL day setup), Mobile responsiveness QA

8.1 Key Build Decisions

- Use Livewire for all interactive components (forms, approvals, bell notifications) — one codebase, no separate frontend repo
- Database queue driver throughout (no Redis) — MySQL handles async notification dispatch at this team size
- Payslip PDFs via barryvdh/laravel-dompdf — single composer package, no external service
- Notification bell uses Livewire polling (every 30 seconds) — simple and reliable without WebSockets
- financial_year column on leave_balances and performance tables stores e.g. '2024-25' to correctly scope to July–June cycle
- salary_cycle (A/B) on employment_details drives which payroll_run an employee belongs to each month
- OT hours on attendance_logs only materialise into overtime_records if a matching approved ot_request exists for that date — enforced in OTService

8.2 Deployment Checklist

32. VPS: Ubuntu 22.04, Nginx, PHP 8.3-FPM, MySQL 8, Composer, Node (Tailwind build)
33. Clone repo, run composer install and npm run build
34. Configure .env: DB credentials, SMTP (payslip email only), APP_KEY, APP_URL
35. php artisan migrate
36. php artisan db:seed — seeds shift_settings, leave_types, public_holidays, december_mandatory_days, departments
37. Add cron entry for Laravel Scheduler
38. Configure Nginx to serve public/ with correct headers
39. Enable HTTPS via Certbot + Let's Encrypt
40. Create first Super Admin: php artisan pulse:create-admin
41. Test in-app notification flow and payslip email end-to-end before go-live

Section 9 — Security

- HTTPS enforced: all traffic over TLS; HTTP redirected at Nginx
- CSRF protection: all Blade forms include @csrf — enabled by default in Laravel
- SQL injection: Eloquent ORM with parameterised queries throughout; no raw string interpolation
- XSS: Blade {{ }} escapes output; {!! !!} used only for trusted admin HTML
- Role enforcement: every controller method checks Laravel Policy; no front-end-only guards
- File uploads: MIME type validated (PDF and images only); stored outside public web root
- Payslip downloads: served via Laravel temporary signed URLs (expire in 5 minutes)
- Password: bcrypt hashing, minimum 8 characters
- Session timeout: 8 hours inactivity; invalidated on logout
- Audit trail: Eloquent model observers log create/update/delete on employees, leave_requests, payslips, salary_structures, ot_requests to an audit_logs table
- Backups: daily MySQL dump via spatie/laravel-backup; retain 30 days
- .env secrets never committed to version control

AUDIT LOG TABLE: audit_logs

Columns: id, user_id, action (created/updated/deleted), model_type, model_id, old_values (JSON), new_values (JSON), ip_address, created_at

Visible to: Super Admin and HR Admin only.

Retention: indefinite (do not prune — required for HR compliance).